

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination – 2023

Course: B. Tech Branch : Electrical Engineering Semester : III

Subject Code & Name: BTEEC303 Electrical & Electronics Measurement

Max Marks: 60

Date: 06-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Draw and Explain Block diagram of generalized instrumentation system.	CO1	6
B) Describe static and dynamic characteristics of measuring instruments.	CO2	6
C) Describe different types of errors occur in electrical measuring instrument.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Compare spring control and gravity control system used in indicating instruments.	CO1	6
B) The inductance of moving iron ammeter with a full scale deflection of 90° at 1.5 amp is given by the expression, $L = 180 + 40\theta - 4\theta^2 - \theta^3 \mu H$, where θ is deflection in radian from zero position. Calculate spring constant.	CO3	6
C) Explain the working of PMMC instrument and derive the equation for damping torque.	CO1	6
Q.3 Solve Any Two of the following.		12
A) With a circuit diagram derive the equation for an unknown self-inductance measurement using HAY's inductance bridge.	CO2	6
B) The power flowing in a 3 phase 3wire balance load system is measured by two wattmeter method. The reading of wattmeter A is 7500W and B is -1500 W then find power factor of the system.	CO3	6
C) Write a short note on Megger.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Draw a block diagram of Integrating type digital voltmeter and explain the function of each block.	CO2	6
B) Explain the characteristics of transducer and sensor.	CO1	6
C) With a suitable sketch explain working of CRO.	CO3	6
Q.5 Solve Any Two of the following.		12
A) Explain the following transducer for measurement of Temperature: 1. Thermocouple 2. RTD	CO1	6
B) Write a short note on Data Acquisition system (DAS).	CO2	6
C) Explain the working of X-Y Recorder.	CO3	6

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